IE 49A

HW1 - An Iterated Prisoner's Dilemma Simulation

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1. Objective and Model

This project investigates how simple behavioral rules can sustain cooperation in the Iterated Prisoner's Dilemma and how mistakes affect their performance. Players repeatedly choose between cooperation (C) and defection (D). The payoff matrix uses temptation T = 5, mutual cooperation reward R = 3, mutual defection punishment P = 1, and the exploited cooperator's payoff S = 0. These values satisfy $T > R > P > S$ and $2R > T + S$.

Although defection offers an immediate individual advantage, mutual cooperation produces a higher collective payoff than mutual defection. Repeated interaction allows players to respond to previous behavior, potentially protecting cooperation through reciprocity.

2. Strategies and Experimental Design

Five strategies were implemented. Always Cooperate (ALL-C) cooperates unconditionally, whereas Always Defect (ALL-D) always defects. Tit-for-Tat (TFT) cooperates initially and then copies the opponent's previous action. Grim Trigger (GRIM) cooperates until the opponent defects once, then defects permanently. Generous Tit-for-Tat (GTFT) follows TFT but cooperates with probability 0.10 after an opponent's defection.

The main experiments compare noiseless and 5% action-noise round-robin tournaments. Each distinct pair of strategies plays 200 rounds, with 30 independent repetitions and random seed 42. Self-play is excluded from these tournaments. Both players choose simultaneously, and their histories reset between matches. Noise independently flips an intended action; players accurately observe the executed actions.

System cooperation is measured as the fraction of executed moves that are cooperative. Each strategy's per-round payoff is averaged across its four opponents, and cumulative payoff sums these averages over rounds. Additional experiments examine pairwise interactions and sensitivity to noise, forgiveness, and the number of rounds.

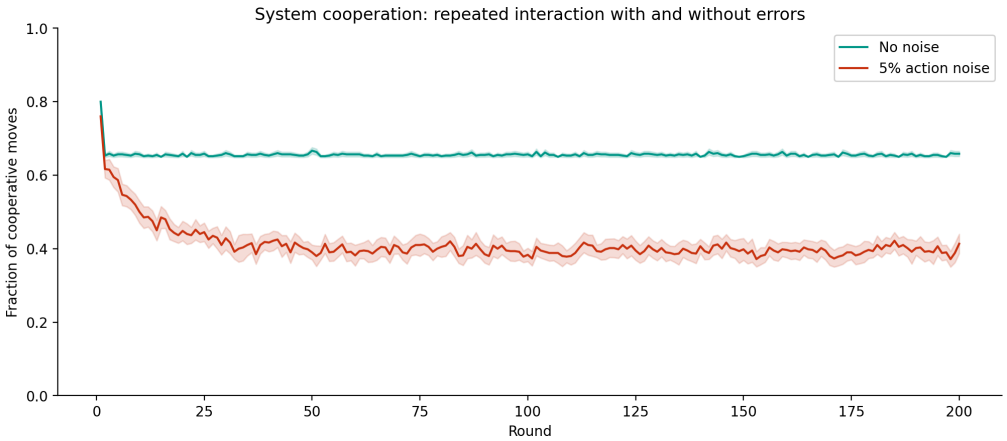


Figure 1. Mean system cooperation over time without noise and with 5% action noise. Shading represents approximately ±2 standard errors across repetitions.

3. Cooperation Results

Overall cooperation was 65.59% without noise and 41.15% with 5% action noise, a decrease of 24.44 percentage points. Cooperation starts relatively high because four strategies initially cooperate. It subsequently declines as players respond to ALL-D or accidental defections. Thus, repeated interaction can sustain cooperation among compatible partners without guaranteeing that system-wide cooperation increases over time.

4. Strategy Performance and Forgiveness

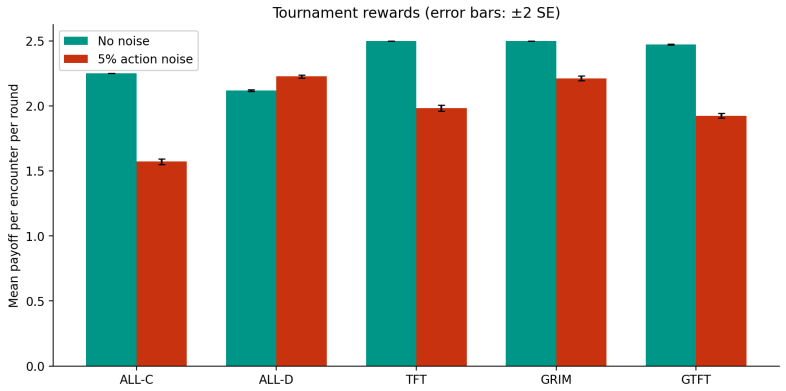


Figure 2. Mean payoff per encounter per round for each strategy. Error bars show approximately ±2 standard errors of repetition-level mean payoffs.

Strategy	No noise	5% action noise
ALL-C	2.250	1.572
ALL-D	2.119	2.227
TFT	2.499	1.983
GRIM	2.499	2.213
GTFT	2.473	1.925

Table 1. Average payoff per encounter per round in the main tournaments.

Without noise, TFT and GRIM achieved the highest average payoff. Both sustain cooperation with responsive opponents while limiting continued exploitation by ALL-D. ALL-C remained vulnerable because it never retaliates. With 5% noise, ALL-D had the highest observed mean payoff, narrowly above GRIM. This descriptive ranking does not establish a statistically significant or universal advantage.

Separate noisy self-pair experiments clarify the benefit of forgiveness. TFT–TFT achieved 51.73% cooperation, compared with 73.03% for GTFT–GTFT. Forgiveness can interrupt retaliation cycles caused by mistakes. However, GTFT earned less than TFT in the mixed tournament, showing that repairing cooperation can also increase vulnerability to persistent defectors.

5. Implementation and Optional Extensions

The self-contained Jupyter notebook includes implementation checks, reproducible experiments, per-round and cumulative payoff plots, payoff heatmaps, and exported figures and tables. Sensitivity analyses explore noise, forgiveness, and interaction length. An interactive dashboard allows users to change strategies, payoffs, rounds, repetitions, probabilities, and the random seed, then update the results through a Run simulation button. These features implement the optional noise and interactive UI extensions.

6. Conclusion and Limitations

The results illustrate how memory and reciprocal responses can sustain cooperation without central coordination. Nevertheless, performance depends on the opponent mix and the reliability of interactions. Forgiveness supports recovery after mistakes but does not guarantee tournament success.

Strategies and encounter frequencies remain fixed, so the model does not demonstrate evolutionary spread or learning. Its conclusions are limited to the tested rules, parameters, and action-noise assumptions. Cooperation is therefore sustained under particular conditions rather than shown to emerge inevitably.